Peristaltic Blood Pump Design Workflow

4-Stage Pipeline with Required Input Data for Optimization

This part of project focuses on the optimization of a peristaltic blood pump designed for hemodialysis extracorporeal circuits. The primary objective of the optimization is to minimize hemodynamic trauma by suppressing transient high-pressure pulses and reducing fluid shear stress during the cyclic compression and decompression of the tube. By integrating analytical kinematics of peristaltic motion with multi-physics numerical simulations, the pump's geometry and operational dynamics were refined to reliably overcome circuit pressure losses while ensuring smooth, continuous, and biocompatible blood transport.

Figure 1 compares the clinical transient pressure response of the hemodialysis pump against the validated optimized suction pressure. The optimized curve effectively smooths out initial start-up oscillations and stabilizes around the nominal operating pressure.

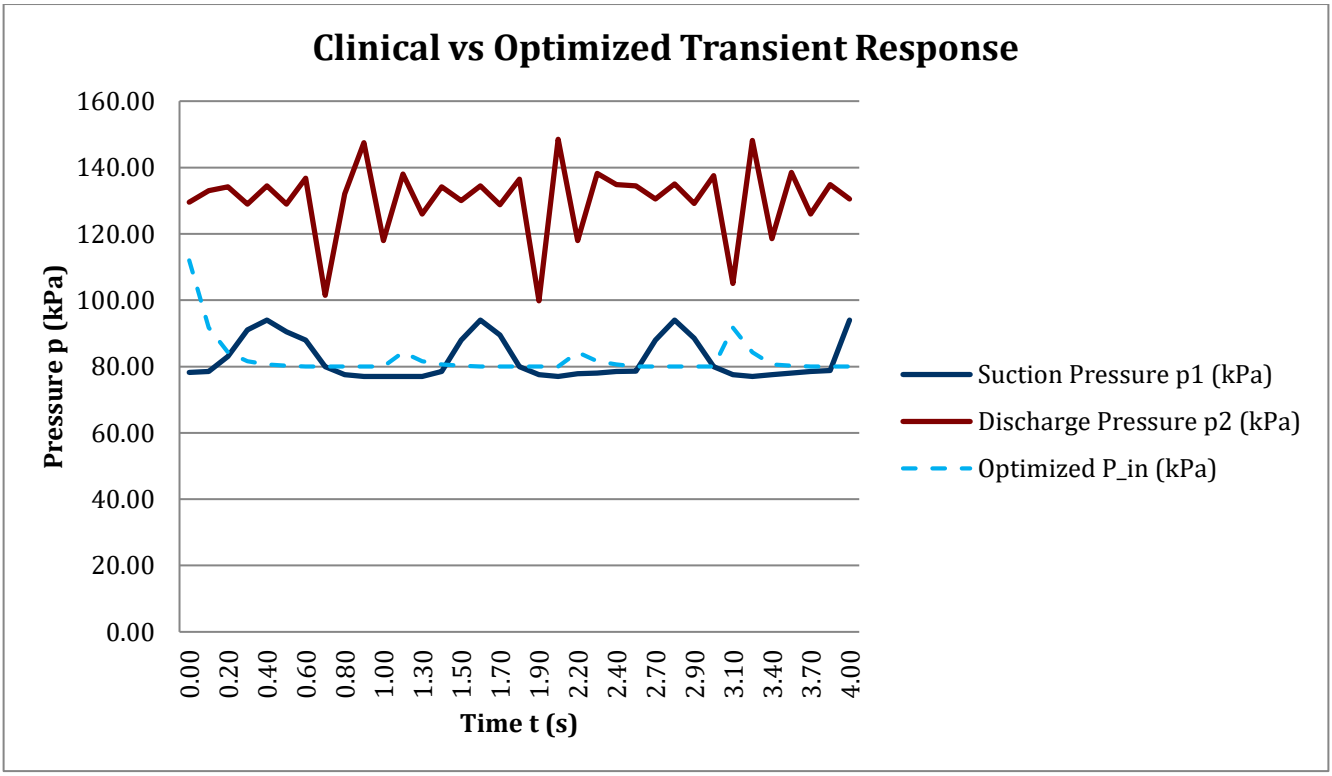


Figure 1: Time-Series Pressure Data

The core objective of this work is the structural and kinematic optimization of a rotary peristaltic blood pump to ensure gentle and efficient extracorporeal blood circulation.

Key Optimization Objectives:

i. **Pressure Pulse Attenuation:** Suppress high-amplitude pressure spikes and backflow reflux during the sudden tube compression and decompression phases.
ii. **Shear Stress & Hemolysis Reduction:** Minimize mechanical shear stress levels imposed on erythrocytes to prevent blood trauma and cell damage.
iii. **Pressure Loss Compensation:** Deliver sufficient head pressure to overcome total dynamic pressure drops across the arterial access, dialyzer filter, and venous return lines.
iv. **Kinematic & Structural Refinement:** Synthesize analytical derivations of peristaltic movement with FEA/CFD co-simulations to optimize casing contours, roller engagement angles, and Tygon tube deformation.

Stage 1: Requirements, Physiological Sizing & Analytical Equations

Parameter	Engineering Inputs & Values	Governing Equation	Primary Design Focus
Flow Rate & System Pressures	• Target Flow (Q): 350 mL/min (5.833×10^{-6} m^3/s) • Suction Pressure (p_1): -13,332 Pa (-100 mmHg) • Venous Return Pressure (p_3): 10,665 Pa (80 mmHg) • Dialyzer Loss (p_dialyzer): 16,665 Pa (125 mmHg) • Discharge Pressure (p_2): 27,331 Pa (205 mmHg) • Differential Pressure (Δp): 40,663 Pa	$p_2 = p_3 + p_{dialyzer} = 27331$ Pa $\Delta p = p_2 - p_1 = 40663$ Pa	Define pressure head requirements across extracorporeal hemodialysis circuit.
Specific Work & Power Output	• Fluid: Blood (ρ = 1050 kg/m^3, ν = 2.65 mm^2/s) • Estimated Efficiency (η_est): 30% • Specific Head (ΔY): 38.73 J/kg • Hydraulic Power Output (P_h): 0.79 W	$\Delta Y = \frac{p_2 - p_1}{\rho} = \frac{27331 - (-13332)}{1050} = 38{,}73 \frac{J}{kg}$ $P_h = \frac{Q \cdot \Delta Y \cdot \rho}{\eta_{est}} = \frac{\frac{350}{60 \cdot 10^6} \cdot 38{,}73 \cdot 1050}{0{,}3} = 0{,}79$ W	Drive sizing estimation for department motor inventory.
Geometric Sizing & Rotational Speed	• Casing Diameter (D): 160 mm • Roller Diameter (d_roller): 30 mm • Number of Rollers (N_roller): 2 • Tube Inner Diameter (d_tube): 13 mm • Wall Thickness (t): 1.5 mm | Occlusion (Oc): 20% • Occlusion Gap (s): 2.4 mm | Outer Dia	$Q_{th} = V_g \cdot n \cdot N_{roller}$ $n = \frac{Q_{th}}{\left(\pi \frac{R + r}{2} - d_{roller}\right) \cdot \left(\frac{\pi d_{tube}^2}{4}\right) \cdot N_{roller}}$	Establish kinematic dimensions and rotational speed.

	(D_tube): 16 mm • Rotor Radius (R_r): 62.5 mm | Length (l): 226.2 mm • Volume (V_g): 17.58 mL | Speed (n): 6.24 RPM	$s = 2t(1 - Oc) = 2.4$ mm	

Stage 2: Hyperelastic Structural Analysis

Step	Technical Specification & Input Values	Governing Equation	Primary Design Focus
Hyperelastic Material Formulation	• Material: Flexible Tygon Tubing • Non-linear strain energy potential model • Experimental Young's Modulus (E): 1.3 MPa • Experimental Shear Modulus (G): 4.0 MPa • Poisson's Ratio (μ): ~0.49 (nearly incompressible)	$W = \frac{1}{2}G(\lambda_1^2 + \lambda_2^2 + \lambda_3^2 - 3)$ $= \frac{1}{2}G(I_1 - 3)$ $G = NkT; \quad G = \frac{E}{2(1+\mu)}$	Accurate stress-strain response modeling under hyperelastic compression.
Boundary Conditions & Mesh Setup	• Fixed support on tube inlet/outlet and casing interface • Displacement boundary condition driven by 30 mm roller motion • Occlusion gap preset to 2.4 mm (20% squeeze)	Displacement Profile: $u_r(t) = f(\theta_{rotor}(t))$ Fixed rim: $u_x = u_y = u_z = 0$	Simulate continuous cyclic tube squeezing without mesh distortion.
Structural Outputs & Life Prediction	• Maximum Von-Mises Stress distribution • Reaction squeezing force on rotor arm • Fatigue life prediction under cyclic deformation	$N_f = C \cdot (\Delta\sigma)^{-\beta}$ $S_{ij} = \frac{\partial W}{\partial E_{ij}}$	Prevent tube spallation, rupture, and excessive shaft deflection.

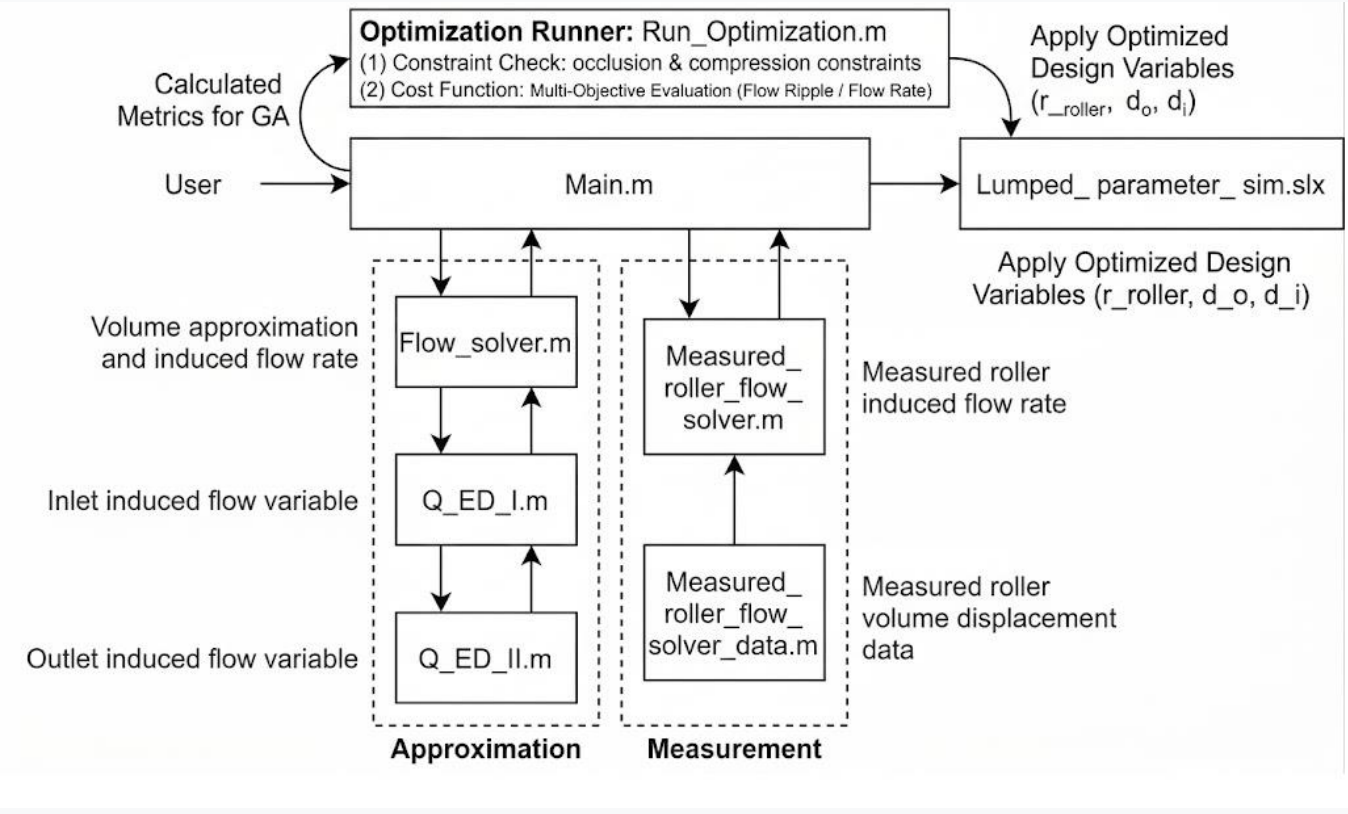
Stage 3: Fluid Dynamics, Shear Stress & Hemolysis Minimization

Parameter	Numerical Setup & Biological Criteria	Governing Equation	Primary Design Focus
Fluid Properties & Navier-Stokes Setup	• Medium: Human Blood • Density (ρ): 1050 kg/m^3 • Kinematic Viscosity (ν): 2.65 mm^2/s • Dynamic Viscosity (μ): ~0.00278 Pa·s • Dynamic / Moving Mesh UDF in ANSYS Fluent	$\rho\left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla\vec{v}\right) = -\nabla P + \mu\nabla^2\vec{v} + \vec{f}$ $\nabla \cdot \vec{v} = 0$	Resolve transient laminar flow field inside moving tube cavity.

Shear Trauma & Pressure Pulse Evaluation	• Evaluation of 4 fluid-mechanical damage mechanisms: 1. Velocity gradients (shear between layers) 2. Pressure variations (cell deformation) 3. Flow stagnation (thrombosis risk) 4. Reverse reflux during disengagement	$\tau = \mu \frac{\partial u}{\partial y}$ $\Delta P_{reflux} = \xi \cdot \frac{\rho v^2}{2}$	Minimize blood trauma and erythrocyte hemolysis.
Elliptic Casing Trajectory Optimization	• Elliptic casing variant to accelerate rollers • Reduces pressure gap in front/behind leading roller • Lower pressure pulse amplitude during tube restitution	$r(\theta) = \frac{ab}{\sqrt{(b\cos\theta)^2 + (a\sin\theta)^2}}$ Pressure Gap $\rightarrow \min(\Delta P_{pulse})$	Eliminate backflow pressure spikes and improve flow smoothness.

Stage 4: System Simulation & Optimization

Subsystems Definition



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============================================================
             PERFORMANCE COMPARISON REPORT
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Design Parameter / Metric | Initial (Baseline) | Optimized
------------------------------------------------------------
Roller Radius (mm)      | 19.8700          | 19.4638
Outer Tube Dia (mm)     | 14.0000          | 15.9374
Inner Tube Dia (mm)     | 10.0000          | 10.8367
Disengage Angle (deg)   | 30.0000          | 35.2993
------------------------------------------------------------
Average Flow (mL/s)     | 77.94            | 89.97
Flow Ripple (mL/s)      | 1.97             | 0.00
Total Cost Score        | -2.0621          | -2.6990
============================================================
Launching Simulink Model with Optimized Parameters: Lumped_parameter_sim.slx ...
Simulink simulation completed successfully.
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Figure 2: Optimization Process

* Stage 4: Model Validation, Limitations, and Real-World Reliability

When assessing the extent to which one can rely 100% on these numerical results in real-world or clinical environments, several key factors must be considered:

1. **Rigid Roller Assumptions:** As discussed, the rollers are treated as completely rigid bodies within this model. Consequently, the mechanical compliance of the rollers themselves and the localized deflection of the housing under high operating pressures are not accounted for.

2. **Actual Elastomeric Tube Behavior:** The model evaluates tube deformation geometrically (through compression ratio and occlusion percentage). However, it does not simulate the complex viscoelastic material properties and hysteresis of the tubing (such as medical-grade silicone) undergoing continuous cyclic compression and release at 170 RPM.

3. **Role of the Simulink Model (`Lumped_parameter_sim.slx`):** This lumped-parameter hydraulic framework—incorporating pressure dynamics, fluid inertia, and system line resistance—provides highly valuable insights into flow pulsations and systemic circuit behavior.

For the preliminary design phase, dimensional optimisation, and comparative performance evaluation (such as our work on maximising flow rate while minimising flow ripple), these outputs are exceptionally robust and provide the correct engineering direction. However, for final prototyping and critical medical device applications (such as blood pumps), these optimised results must ultimately be validated through physical bench testing on the actual hardware.

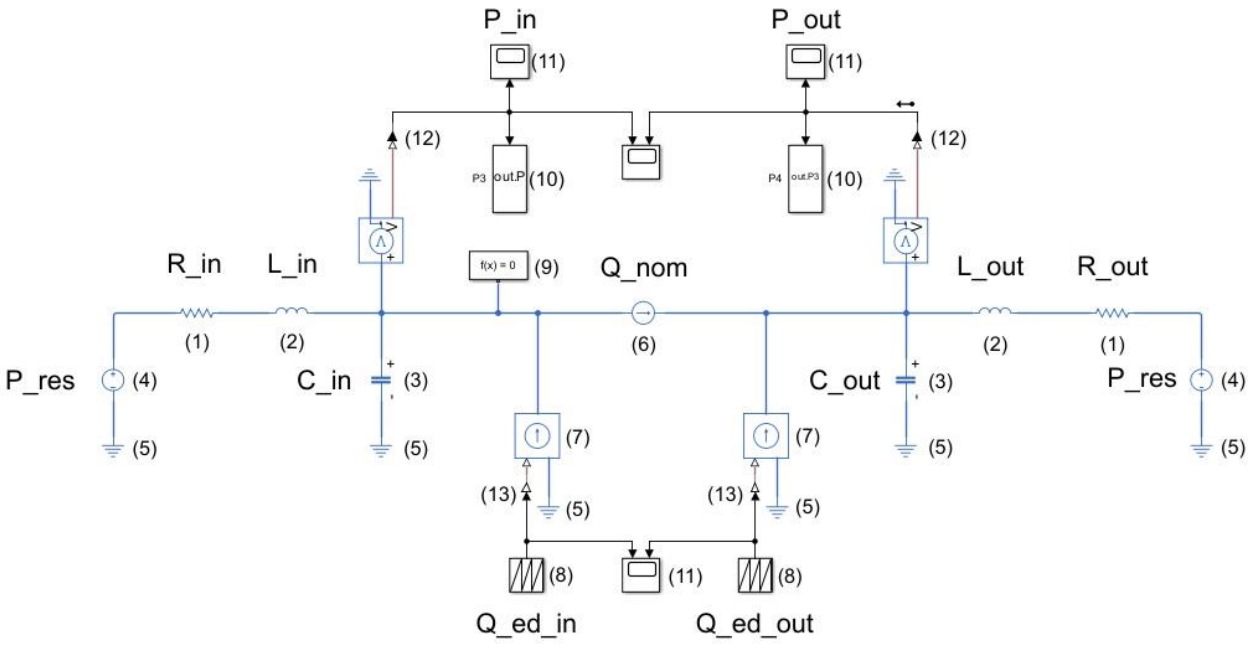


Figure 3: Lumped-parameter modeling of a peristaltic pump in Simulink

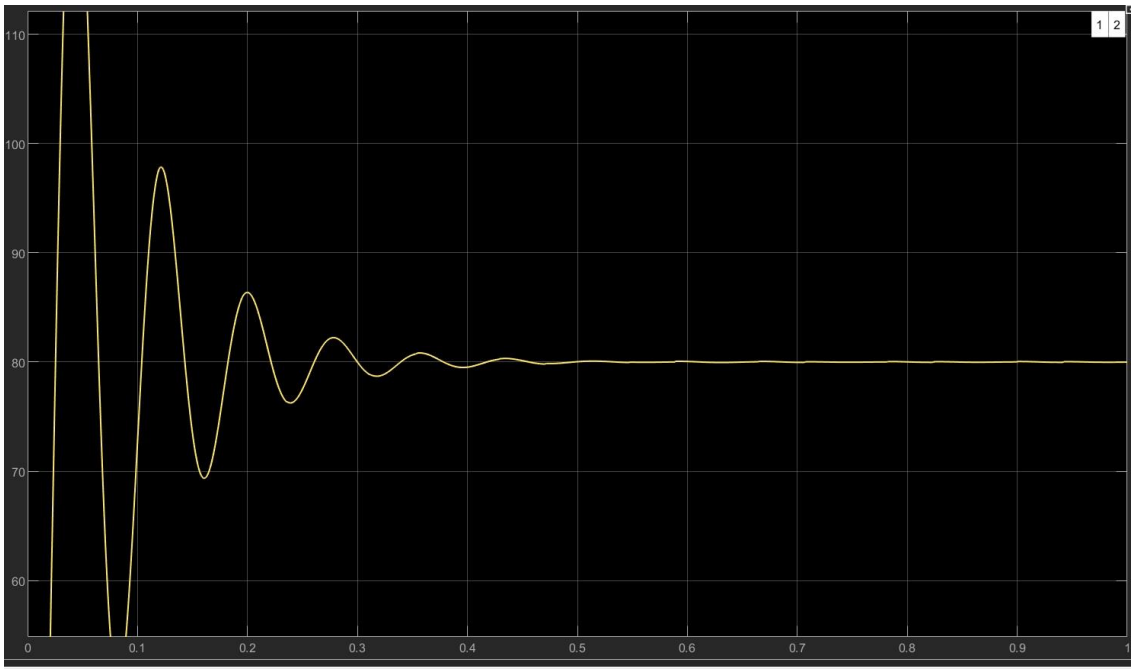


Figure 4: Suction Pressure Ressult from Simulink